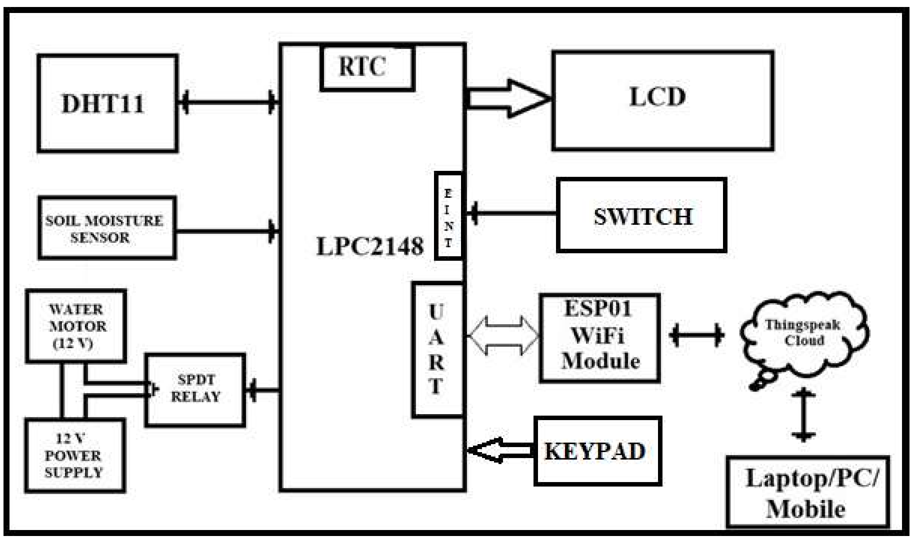
SMART IOT IRRIGATION SYSTEM

AIM:

The main aim of the project is to monitor the irrigation system over IoT. For that read the environmental parameter values (Ex. Temperature, Relative Humidity) and update that information to thingspeak cloud. And with the help of soil moisture sensor information automatic on/off control applied on water pump. Hence, will automatically irrigate the field.

Block diagram:



HARDWRAE REQUIREMENTS:

- LPC2148
- DHT11 SENSOR
- LCD
- WI-FI MODULE (ESP01)
- SOIL MOISTURE SENSOR
- DB-9 CABLE/USB-UART CONVERTER

SOFTWARE REQUIREMENTS:

- KEIL C Compiler
- PROGRAMMING IN EMBEDDED C
- Flash Magic

Steps to be followed to complete your project:

- Create New Folder in your laptop/PC and save that folder with your project name.
- Then copy what you have done - lcd.c, lcd.h, delay,c, delay.h, uart.c & uart.h, into project folder.
- Individuals can check each module.
- First check lcd to display character constant, string constant and integer constant.
- Next Check UART peripheral by transmitting string constant on hyper terminal.
- Then download the uart interrupt code (from LMS) and check the working on hardware board.
- Next connect DHT11 sensor to LPC2148 and develop the driver for DHT11 (reading temperature and humidity from DHT11) and display on LCD.
- Next check the working condition of ESP01 with the help of flash magic terminal. (Which commands are required to use for your project refer to the screen shots under reference data folder)
- Next connect the ESP01 module to LPC2148. (You can refer to the data sheet of ESP01 to know how to connect)
- Then develop the driver for ESP01 for LPC2148. By using ESP01 driver, send one constant data to thingspeak cloud.
- Next check the soil moisture sensor working condition. For that connect the soil moisture sensor to any one of the GPIO pin and read the status of soil moisture sensor output. Based on that need to control the water motor. Soil moisture sensor gives the result is two forms. One is digital (0/1) and other one is analog. If users want to check the soil moisture percentage, then it is preferable to use analog.
- Whenever the sensor detects a low quantity of moisture in the soil, then

turn ON the motor. Hence, it will automatically irrigate the field. Once the soil becomes wet, then turn OFF the motor. Based on the requirement needed to control the motor.

- After checking DHT11, ESP01 driver and soil moisture sensor then implement the main logic. Inside main initialize all required peripherals.
- If user wants to change the RTC information, it needs to generate interrupt signal by pressing switch which is connected to interrupt pin of the LPC2148. Users need to change the required timing information by following the menu-based time editing process through 4x4 matrix keypad. (Note: add all required test cases).
- In continuous loop, read current temperature value and relative humidity values from DHT11 and display it on LCD. Then send the same information to ThingSpeak cloud with respect to some time-gap (use RTC information for time gap monitoring). For irrigation control, soil moisture should be considered as the primary parameter. Temperature can be used as an additional parameter to make the system smarter. If the soil moisture level is low, then turn ON the water motor. If the soil is dry and the temperature is high, the motor can be kept ON for a longer duration (3 Min). If the soil is dry but temperature is normal/low, the motor can be turned ON for a shorter duration (1 Min). After motor is ON, send that information to the cloud. After motor is OFF, that information is also sent to the cloud.
- If you’re getting this output, then your project is completed.

Note: Use LED instead of water motor to show the result.

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DIFFERENT IOT APPLICATION PLATFORMS: (FYR)

AWS IoT: Amazon Web Services (AWS) IoT is a cloud-based platform that allows users to connect and manage IoT devices and applications. It provides a wide range of services for data collection, storage, analysis, and visualization.

Microsoft Azure IoT: Microsoft Azure IoT is a comprehensive cloud-based platform that provides tools and services for IoT device management, data analysis, and application development. It supports a wide range of devices and platforms, including Windows and Linux.

Google Cloud IoT: Google Cloud IoT is a cloud-based platform that provides services for IoT device management, data processing, and application development. It supports a wide range of devices and platforms, including Android and iOS.

IBM Watson IoT Platform: IBM Watson IoT Platform is a cloud-based platform that provides services for IoT device management, data analysis, and application development. It supports a wide range of devices and platforms, including Raspberry Pi and Arduino.

Thing Speak: Thing Speak is an open-source Internet of Things (IoT) application platform and API (Application Programming Interface) that allows users to collect, analyse and act on data from various sources. It was developed by MathWorks, a leading provider of technical computing software. Thing Speak provides an easy-to-use interface for users to create and manage IoT devices and applications. It supports a wide range of data sources, including sensors, web services, and social media. The platform also includes features for data visualization, data analysis, and real-time alerts.

Cayenne: Cayenne is a cloud-based IoT platform that provides services for device management, data visualization, and automation. It supports a wide range of devices and platforms, including Arduino and Raspberry Pi.

Kaa IoT Platform: Kaa IoT Platform is an open-source IoT platform that provides tools and services for IoT device management, data processing, and application development. It supports a wide range of devices and platforms, including Android and iOS.

Ubidots: Ubidots is a cloud-based IoT platform that provides services for data collection, visualization, and analysis. It supports a wide range of devices and platforms, including Arduino and Raspberry Pi.

Losant: Losant is a cloud-based IoT platform that provides services for device management, data visualization, and automation. It supports a wide range of devices and platforms, including MQTT and HTTP.

Xively: Xively is a cloud-based IoT platform that provides services for device management, data analytics, and application development. It supports a wide range of devices and platforms, including Arduino and Raspberry Pi.

Blynk: Blynk is a mobile app and cloud-based IoT platform that provides services for device management, data visualization, and control. It supports a wide range of devices and platforms, including Arduino and Raspberry Pi.

Particle: Particle is an IoT platform that provides tools and services for device management, data visualization, and firmware development. It supports a wide range of devices and platforms, including Wi-Fi and cellular.

Thinger.io: Thinger.io is a cloud-based IoT platform that provides services for device management, data collection, and analysis. It supports a wide range of devices and platforms, including ESP8266 and ESP32.

DeviceHive: DeviceHive is an open-source IoT platform that provides services for device management, data analytics, and application development. It supports a wide range of devices and platforms, including Java and .NET.

Mongoose OS: Mongoose OS is an open-source IoT platform that provides services for device management, data analytics, and firmware development. It supports a wide range of devices and platforms, including ESP32 and STM32.

Hologram: Hologram is a cloud-based IoT platform that provides services for device management, data collection, and visualization. It supports a wide range of devices and platforms, including Raspberry Pi and BeagleBone.

ioBroker: ioBroker is an open-source IoT platform that provides services for device management, data visualization, and automation. It supports a wide range of devices and platforms, including Node.js and Docker.

Cayenne: Cayenne is a cloud-based IoT platform that provides services for device management, data visualization, and automation. It supports a wide range of devices and platforms, including Arduino and Raspberry Pi.

Altair SmartWorks: Altair SmartWorks is a cloud-based IoT platform that provides services for device management, data analytics, and application development. It supports a wide range of devices and platforms, including industrial sensors and machines.

Helium: Helium is a decentralized IoT platform that provides services for device management, data collection, and application development. It supports a wide range of devices and platforms, including